

```
pranav_chowdary@Pranav: ~  
GNU nano 8.7.1 q5a.c  
#include <stdio.h>  
#include <unistd.h>  
#include <string.h>  
#include <sys/wait.h>  
#include <time.h>  
  
int main()  
{  
    int fd[2];  
    char msg[] = "Hello from Parent";  
    char buffer[100];  
  
    pipe(fd);  
    clock_t start = clock();  
    int pid = fork();  
    if (pid == 0)  
    {  
        close(fd[1]);  
        read(fd[0], buffer, sizeof(buffer));  
        printf("Child consumed: %s\n", buffer);  
        close(fd[0]);  
    }  
    else  
    {  
        close(fd[0]);  
        printf("Parent produced: %s\n", msg);  
        write(fd[1], msg, strlen(msg) + 1);  
        close(fd[1]);  
        wait(NULL);  
    }  
}
```

Read 49 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was

```
pranav_chowdary@Pranav: ~  
GNU nano 8.7.1 q5a.c  
char buffer[100];  
pipe(fd);  
clock_t start = clock();  
int pid = fork();  
if (pid == 0)  
{  
    close(fd[1]);  
    read(fd[0], buffer, sizeof(buffer));  
    printf("Child consumed: %s\n", buffer);  
    close(fd[0]);  
}  
else  
{  
    close(fd[0]);  
    printf("Parent produced: %s\n", msg);  
    write(fd[1], msg, strlen(msg) + 1);  
    close(fd[1]);  
    wait(NULL);  
    clock_t end = clock();  
    double time = (double)(end - start) / CLOCKS_PER_SEC;  
    printf("Communication time: %f seconds\n", time);  
}  
return 0;  
}
```

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was

```
pranav_chowdary@Pranav: ~  
pranav_chowdary@Pranav:~$ nano q5a.c  
pranav_chowdary@Pranav:~$ gcc q5a.c -o q5a  
./q5a  
Parent produced: Hello from Parent  
Child consumed: Hello from Parent  
Communication time: 0.000189 seconds  
pranav_chowdary@Pranav:~$ |
```

B

```
GNU nano 8.7.1 q5b.c  
#include <stdio.h>  
#include <unistd.h>  
#include <sys/wait.h>  
  
int main()  
{  
    int fd[2];  
    pipe(fd);  
  
    int pid1 = fork();  
    if (pid1 == 0)  
    {  
        dup2(fd[1], STDOUT_FILENO);  
        close(fd[0]);  
        close(fd[1]);  
        execlp("ls", "ls", "-l", NULL);  
    }  
  
    int pid2 = fork();  
    if (pid2 == 0)  
    {  
        dup2(fd[0], STDIN_FILENO);  
        close(fd[1]);  
        close(fd[0]);  
        execlp("grep", "grep", ".c", NULL);  
    }  
  
    close(fd[0]);  
    close(fd[1]);  
    wait(NULL);  
    wait(NULL);  
}
```

Read 42 lines

⌘ Help	⌘ Write Out	⌘ Where Is	⌘ Cut	⌘ Execute	⌘ Location	⌘ Undo	⌘ Set Mark	⌘ To Bracket
⌘ Exit	⌘ Read File	⌘ Replace	⌘ Paste	⌘ Justify	⌘ Go To Line	⌘ Redo	⌘ Copy	⌘ Where Was

```
pranav_chowdary@Pranav: ~$ nano q5b.c
GNU nano 8.7.1 q5b.c

int main()
{
    int fd[2];
    pipe(fd);

    int pid1 = fork();
    if (pid1 == 0)
    {
        dup2(fd[1], STDOUT_FILENO);
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
    }

    int pid2 = fork();
    if (pid2 == 0)
    {
        dup2(fd[0], STDIN_FILENO);
        close(fd[1]);
        close(fd[0]);
        execlp("grep", "grep", ".c", NULL);
    }

    close(fd[0]);
    close(fd[1]);

    wait(NULL);
    wait(NULL);

    return 0;
}
```

```
pranav_chowdary@Pranav: ~$ nano q5b.c
pranav_chowdary@Pranav: ~$ gcc q5b.c -o q5b
a.out copy.txt main.c p1.c q1 q1.c q2 q2.c q3 q3.c q4 q4.c q5a q5a.c q5b q5b.c source.txt t2.txt zombie zombie.c
pranav_chowdary@Pranav: ~$ ./q5b
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16120 Aug 25 05:44 a.out
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 44 Aug 22 08:45 copy.txt
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 60 Aug 17 04:03 main.c
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 246 Aug 25 05:43 p1.c
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16264 Aug 22 08:45 q1
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 426 Aug 22 08:39 q1.c
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16168 Aug 22 08:45 q2
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 365 Aug 22 08:40 q2.c
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16208 Aug 22 08:45 q3
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 478 Aug 22 08:43 q3.c
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16160 Aug 22 08:45 q4
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 450 Aug 22 08:44 q4.c
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16336 Sep 3 05:07 q5a
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 777 Sep 3 04:52 q5a.c
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16208 Sep 3 05:08 q5b
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 551 Sep 3 04:57 q5b.c
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 44 Aug 22 08:41 source.txt
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 9 Aug 25 05:44 t2.txt
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16168 Aug 22 08:45 zombie
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 365 Aug 22 08:44 zombie.c
pranav_chowdary@Pranav: ~$ ls -l | grep ".c"
-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16120 Aug 25 05:44 a.out
-rw-r--r-- 1 pranav_chowdary pranav_chowdary 44 Aug 22 08:45 copy.txt
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-rwxr-xr-x 1 pranav_chowdary pranav_chowdary 16264 Aug 22 08:45 q1
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-rw-r--r-- 1 pranav_chowdary pranav_chowdary 9 Aug 25 05:44 t2.txt
```